Differential Geometry

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Lecture 1

1 Manifolds

The concept of a manifold is intended to generalise curves and surfaces in $\mathbb{R}^3$.

Definition 1.1 (smooth structure)

A *smooth structure* on a topological space M is a collection of *charts* (an *atlas*)

$$\varphi_\alpha : U_\alpha \to V_\alpha$$

where $U_\alpha \subseteq M$, $V_\alpha \subseteq \mathbb{R}^d$ are open, φ_α is a homeomorphism, such that

1. $M = \bigcup_\alpha U_\alpha$. That is, the charts cover M.
2. Given any charts $\varphi_\alpha, \varphi_\beta$, the map
$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$
is a smooth map between open subsets of $\mathbb{R}^d$.
3. If φ is *compatible* (as in the previous point) with every chart in the atlas, then φ is a chart in the atlas.

In practice (i.e. using Zorn's lemma), given any collection of charts satisfying the first two conditions, we can find a maximal atlas containing it.

Remark 1.2.
1. The second condition implies that $\varphi_\beta \circ \varphi_\alpha^{-1}$ is a diffeomorphism.
2. d is fixed, assuming that M is connected.

Definition 1.3 (manifold)

A *(smooth) manifold* M is a second countable Huasdorff space topological space, equipped with a smooth structure.

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$d = \dim(M)$ is the *dimension* of M.

In practice, we may induce a topology on M from a smooth structure (without the mention of open or homeomorphism), then $D \subseteq M$ is open if for any chart φ_α, $\varphi_\alpha(D \cap U_\alpha)$ is an open subseteq of $\mathbb{R}^d$.

Remark 1.4.
1. We can also define a C^k manifold by replacing smooth with C^k as above. If $k = 0$, then we have a *topological manifold*.
2. On the other hand, we can replace $\mathbb{R}^d$ with $\mathbb{C}^n$, and obtain a *complex manifold*.

$\mathbb{R}^n$ is (trivially) a manifold.

Example 1.5 (unit sphere)
Define

$$S^n = \{x = (x_0, \dots, x_n) \in \mathbb{R}^n \mid ||x|| = 1\} \subseteq \mathbb{R}^{n+1}$$

For the charts, we can consider the stereographic projections

$$\varphi(x) = \frac{1}{1 - x_0}(x_1, \dots, x_n) \qquad \text{for } x_0 \neq 1$$

and

$$\psi(x) = \frac{1}{1 + x_0}(x_1, \dots, x_n) \qquad \text{for } x_0 \neq -1$$

In particular,

$$\psi \circ \varphi^{-1}(u) = \frac{u}{||u||^2}$$

Example 1.6
If M, N are manifolds of dimensions d, e respectively, then $M \times N$ is a manifold of dimension $d + e$.

As a corollary, $T^n = (S^1)^n$ is a manifold.

Example 1.7
If M is a manifold, $X \subseteq M$ is open, then X is a manifold.

Lecture 2

A convention, we will will assume for any manifold M, we will assume all components are of the same dimension. In particular, $\dim(M)$ is well defined.

Notation 1.8. We will write M^d to say M is a manifold of dimension d.

Example 1.9 (real projective space)
The real projective space is

$$\mathbb{RP}^n = \{\text{straight lines in } \mathbb{R}^{n+1} \text{ through } 0\}$$

Points in $\mathbb{RP}^n$ can be written using homogeneous coordinates $(x_0 : \cdots : x_n)$, where the x_i are not all zero, and

$$(x_0 : \cdots : x_n) = (\lambda x_0 : \cdots : \lambda x_n)$$

for all $\lambda \neq 0$. We will induce a topology from a smooth structure. The charts are given by (φ_i, U_i) where

$$U_i = \{x_i \neq 0\}$$

and we define

$$\varphi_i(x_0 : \cdots : x_n) = \left(\frac{x_0}{x_i}, \ldots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \ldots, \frac{x_n}{x_i} \right) \in \mathbb{R}^n$$

For $i < j$,

$$\varphi_j \circ \varphi_i^{-1}(y_0, \ldots, y_n) = \left(\frac{y_1}{y_j}, \ldots, \frac{1}{y_j}, \ldots, \frac{y_n}{y_j} \right)$$

where the 1 is in the i-th position, and we omit the j-th position. Thus, $\mathbb{RP}^n$ is an n-dimensional manifold. Similarly, we can define the complex projective space $\mathbb{CP}^n$, which is a $2n$-manifold (and a n-dimensional complex manifold).

Example 1.10 (Grassmannian)

$$\mathrm{Gr}(k, n) = \{k\text{-dimensional subspaces of } \mathbb{R}^n\}$$

Then $\mathrm{Gr}(k, n)$ is a manifold of dimension $k(n-k)$. Note we can also define the Grassmannian over $\mathbb{C}$.

Sketch proof. An example of a coordinate neighbourhood is

$$U = \left\{k\text{-dimensional subspaces obtainable as the span of rows of } k \times n \text{ matrices of the form } \begin{pmatrix} I_k & * \end{pmatrix}\right\}$$

where I_k is the $k \times k$ identity matrix. In this case, the $k \times (n-k)$ block $*$ defines local coordinates. Choosing different columns, we have $\binom{n}{k}$ neighbourhoods, $U_{i_1,\ldots,i_k}$, where $i_1 < i_2 < \cdots < i_k$. □

In particular, note that $\mathbb{RP}^n = \mathrm{Gr}(1, n+1)$.

Example 1.11 (Non-example)

Consider an equivalence relation on $\mathbb{R}^2$ given by

$$(x, y) \sim \left(\lambda x, \frac{1}{\lambda} y \right)$$

for any $\lambda \neq 0$. Let $X = \mathbb{R}^2/\sim$ for the quotient, with the quotient topology. We can consider

$$\{(x, y) \mid xy = c\}$$

If $c \neq 0$, then this is one equivalence class. If $c = 0$, this splits into three, which are

- $\{(0, 0)\}$,
- $\{(x, 0) \mid x \neq 0\}$,
- $\{(0, y) \mid y \neq 0\}$

We can write

$$X = (-\infty, 0) \cup (0, \infty) \cup \{0', 0'', 0'''\}$$

We have three charts φ_i corresponding to each choice of zero. For example,

$$U_1 = (-\infty, 0) \cup \{0'\} \cup (0, \infty)$$

with the "natural" map φ_1. On the other hand, the induced topology is not Hausdorff. Note the induced topology is *not* the quotient topology.

Refer to Examples Sheet 1 Question 12 for an example of a non-second countable (non-)example.

Definition 1.12 (smooth map)

Let M, N be manifolds, a continuous map $f : M \to N$ is *smooth* if for any $p \in M$, we can find charts

(φ, U) near p, (ψ, V) near $f(p)$, and the composition

$$\psi \circ f \circ \varphi^{-1}$$

is smooth. as a map on

$$\varphi(U \cap f^{-1}(V))$$

Definition 1.13 (diffeomorphism)
A smooth map $f : M \to N$ is a *diffeomorphism* if f is a bijection, and f^{-1} is smooth. In this case, we say that M, N are *diffeomorphic*.

Proposition 1.14.
1. If $U \subseteq \mathbb{R}^n$, then $f : U \to \mathbb{R}^m$ is smooth in the above definition if and only if it is smooth in the sense of multi-variate calculus.
2. Every chart $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^d$ is a smooth map of manifolds. Moreover, it is a diffeomorphism.
3. If $f : M \to N$, $g : N \to P$ are smooth, then $g \circ f : M \to P$ is smooth.

Proof. Obvious, and so omitted. □

1.1 Matrix Lie groups

Consider the group $\mathrm{GL}(n, \mathbb{R}) \subseteq \mathbb{R}^{n^2}$. In fact, this is an open subset, and so it is a manifold. Moreover, matrix multiplication and inversion are smooth.

Definition 1.15 (Lie group)
A group G is called a *Lie group* if

- G is a manifold,
- the multiplication map $G \times G \to G$ is smooth,
- inversion $G \to G$ is smooth.

Remark 1.16. Suffices to show that the map

$$G \times G \ni (\sigma, \tau) \mapsto \sigma\tau^{-1} \in G$$

is smooth.

In particular, $\mathrm{GL}(n, \mathbb{R}$ is a Lie group. Similarly, $\mathrm{GL}(n, \mathbb{C})$ is also a Lie group. More generally, for $A = (a_{ij}) \in \mathrm{Mat}_n(\mathbb{C})$, we define the norm $\|A\| = n \max_{i,j} |i, j|$. (Note in finite dimensional, all norms are equivalent). In particular,

$$\|AB\| \leq \|A\|\|B\|$$

We define the *exponential of* A as

$$\exp(A) = \sum_k \frac{A^k}{k!}$$

This series converges for all A, by the same proof as for $\exp : \mathbb{C} \to \mathbb{C}$ (or more generally, any Banach algebra). Moreover, it converges uniformly on balls

$$\{\|A\| \leq R\}$$

by the Weierstrass M-test. In particular, $\exp : \mathrm{Mat}_n(\mathbb{C}) \to \mathrm{Mat}_n(\mathbb{C})$ is continuous.

Lecture 3

First of all, note that

$$f(A) = A^n$$

is a differentiable function, with

$$\left\|(\mathrm{d}f)_A\right\| \leq n|A|^{n-1}$$

Therefore, the series which we get by differentiating term by term, converges uniformly on any bounded set, again by the Weierstrass M-test.

Therefore, $\exp(A)$ is (at least) C^1 on $\mathrm{Mat}_n(\mathbb{C})$. We can similarly consider the higher derivatives, and we find that $\exp(A)$ is C^∞.

Remark 1.17. Some basic properties of the matrix exponential:

1. $\exp\left(A^{\mathsf{T}}\right) = \exp(A)^{\mathsf{T}}$,
2. $\exp(A^*) = \exp(A^*)$,
3. $\exp\left(CAC^{-1}\right) = C\exp(A)C^{-1}$,

In general,

$$\exp(A+B) = \exp(A)\exp(B)$$

does not hold. However, it does hold if $AB = BA$. In particular,

4. $\exp(A)\exp(-A) = \exp(-A)\exp(A) = I$

We can also define the *matrix logarithm* via the series

$$\log(I+A) = \sum_m (-1)^{m+1}\frac{A^m}{m}$$

We can show that this converges absolutely and uniformly on $\{|A| < \varepsilon\}$ for any $0 < \varepsilon < 1$. The same result holds for the term by term differentiation, and again we get that $\log(I+A)$ is C^∞ on $\{|A| < 1\}$

Suppose $\log(A)$ is defined, that is, $|A - I| < 1$, then

$$\exp(\log(A)) = A$$

since the same proof as in $\mathbb{C}$ works for any Banach algebra.

On the other hand, if we would like

$$\log(\exp(A)) = A$$

we clearly need $|\exp(A) - I| < 1$. Suppose

$$A_\theta = \begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix}$$

for some $\theta \in \mathbb{R}$. In particular,

$$A_\theta^2 = -\theta^2 I$$

and we find that

$$\exp(A_\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$$

Setting $\theta = 2\pi$, then we find that $\exp(A_{2\pi}) = I$, but

$$\log(\exp(A_{2\pi})) = \log(I) = 0 \neq A_{2\pi}$$

Hence $|\exp(A) - I| < 1$ is a necessary, but not sufficient. On the other hand, $|A| < \log(2)$ is sufficient, since in this case,

$$\left|e^{|A|} - 1\right| < 1$$

which is required for absolute convergence.

Example 1.18 (orthogonal group)

Consider

$$O(n) = \left\{A \in \mathrm{GL}(n,\mathbb{R}) \mid AA^{\mathsf{T}} = A^{\mathsf{T}}A = I\right\}$$

We will show that $O(n)$ is a Lie group. Let A be an orthogonal matrix, $|A - I| < \varepsilon < 1$, then $B = \log(A)$ is defined. Moreover, $\exp(B) = A$. By continuity, we can take ε small enough such that $|B| < \log(2)$. Now

$$\exp(B)\exp\left(B^{\mathsf{T}}\right) = AA^{\mathsf{T}} = \mathrm{id}$$

and so

$$\exp(B) = A = (A^{\mathsf{T}})^{-1} = \exp\left(B^{\mathsf{T}}\right)^{-1} = \exp\left(-B^{\mathsf{T}}\right)$$

Taking log, since $|B| = \left|B^{\mathsf{T}}\right| < \log(2)$, we see that $B = -B^{\mathsf{T}}$.

Conversely, if B is skew-symmetric, with $|B| < \log(2)$, then

$$\exp(B)\exp\left(B^{\mathsf{T}}\right) = I$$

Define

$$V_0 = \left\{B \in \mathrm{Mat}_n(\mathbb{R}) \mid |B| < \log(2), B^{\mathsf{T}} + B = 0\right\}$$

and define

$$U = \exp(V_0)$$

which is an open neighbourhood of $I \in O(n)$.

Proposition 1.19. *$O(n)$ has a C^∞ structure, making it into a manifold and a Lie group, with dimension $n(n-1)/2$.*

Proof. We will use the subspace topology induced from $O(n) \subseteq \mathbb{R}^{n^2}$. Define

$$\begin{aligned} h : U &\to V_0 \\ h(A) &= \log(A) \end{aligned}$$

This is a homeomorphism onto its image, which is an open neighbourhood of $0 \in \mathfrak{so}(n) \cong \mathbb{R}^{n(n-1)/2}$.

Given $C \in O(n)$, define

$$U_C = \{CA \mid A \in U\}$$

and

$$\begin{aligned} h_C : U_C &\to V_0 \\ h_C(A) &= \log\left(C^{-1}A\right) \end{aligned}$$

Again, this is a homeomorphism onto its image. Since $C \in U_C$, we have an open cover. Moreover,

$$h_{C_2} \circ h_{C_1}^{-1}(B) = h_{C_2}(C_1\exp(B)) = \log\left(C_2^{-1}C_2\exp(B)\right)$$

which is a smooth map between open neighbourhoods of $\mathfrak{so}(n)$. Therefore, $O(n)$ is a smooth manifold. Similarly, for consider the map

$$F(A_1, A_2) = A_1A_2^{-1}$$

In local coordinates,

$$\begin{aligned} F_{\mathrm{loc}}(B_1, B_2) &= h_{C_1C_2^{-1}}(F(h_{C_1}^{-1}(B_1), h_{C_2}^{-1}(B_2))) \\ &= \log\left((C_1C_2^{-1})^{-1}C_1\exp(B_1)(C_2\exp(B_2))^{-1}\right) \\ &= \log\left(C_2\exp(B_1)\exp(-B_2)C_2^{-1}\right) \end{aligned}$$

and we can see that this is a smooth map as a function of B_1, B_2. □

This method works for more groups, and is sometimes called the Cayley construction (for matrix Lie groups).

2 Tangent spaces to manifolds

Consider a local curve in a ball in $\mathbb{R}^n$, defined by a smooth parametrisation $x(t) = (x_1(t), \ldots, x_n(t))$. Suppose we have that $x(0) = p \in \mathbb{R}^n$. In this case, its tangent vector at p is

$$\dot{x}(0) \in \mathbb{R}^n$$

If $y(x)$ is a smooth change of coordinates near p, then by the chain rule, we have

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} y(x(t)) = J(y)_p \dot{x}(0)$$

where $J(y)$ is the Jacobian matrix. More explicitly,

$$\dot{y}_i(0) = \sum_j \frac{\partial y_j}{\partial x_j} \dot{x}_j(0)$$

Definition 2.1 (tangent vector)

A *tangent vector* a to a manifold M^n at a point $p \in M$, is an assignment to each coordinate chart (U, φ) with $p \in U$, an n-tuple of coordinates $(a_1, \ldots, a_n) \in \mathbb{R}^n$, such that if (U', φ') is another chart near p, $(x_i), (x_i')$ the respective local coordinates,

$$a_i' = \sum_j \frac{\partial x_i'}{\partial x_j} a_j$$

This definition is called the *tensorial definition*, where we focus on the transformation law. There are several other definitions, for example, derivations, equivalence of curves to first order and so on.

Definition 2.2 (tangent space)

The *tangent space* to M at a point p, denoted $\mathrm{T}_p M$ is the space of all tangent vectors to M at p. It is naturally a vector space (of dimension n over $\mathbb{R}$).

Lecture 4

A choice of chart (with local coordinates (x_i) say), determines a linear isomorphism

$$\mathrm{T}_p M \to \mathbb{R}^n$$

where $n = \dim(M)$. We denote by $\frac{\partial}{\partial x_i}$, the basis corrsponding to the usual standard basis of $\mathbb{R}^n$, under this linear isomorphism.

If we want to emphasise the point $p \in M$, we may write

$$\left(\frac{\partial}{\partial x_i}\right)_p \text{ or } \left.\frac{\partial}{\partial x_i}\right|_p$$

Then by the usual linear algebra, we have that

$$\left(\frac{\partial}{\partial x_i}\right)_p = \sum_j \left(\frac{\partial x_j'}{\partial x_i}\right)_p \left(\frac{\partial}{\partial x_j'}\right)_p$$

An alternative view on tangent vectors is as *derivations*. Given

$$a = \sum a_i \frac{\partial}{\partial x_i} \in \mathrm{T}_p M$$

we can define a *first-order derivation* at p,

$$a : C^\infty(M) \to \mathbb{R}$$

where

$$a(f) = \sum_i a_i \frac{\partial f}{\partial x_i}$$

in local coordinates. This definition is independent of the choice of local coordinates. We can define this in a coordinate independent way, by defining

$$a(f) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} f(x(t))$$

where x is a local curve on M, with $\dot{x}(0) = a$. Moreover, every derivation satisfies the *Leibniz rule*

$$a(fg) = f \cdot a(g) + a(f) \cdot g$$

Conversely, every linear map $a : C^\infty(M) \to \mathbb{R}$ satisfying the Leibniz rule above arises as a derivation corresponding to a tangent vector.

Example 2.3

Let $r : \mathbb{R}^2_{u,v} \supseteq D \to S \subseteq \mathbb{R}^3$ be a regular parametrisation for a surface S. Then $\varphi = r^{-1}$ defines a chart. The partial derivatives r_u, r_v correspond to

$$\frac{\partial}{\partial u}, \frac{\partial}{\partial v}$$

respectively.

For a Lie group G, the tangent spaces has an "infinitessimal counterpart" of the group structure.

Definition 2.4 (Lie algebra)

A *Lie algebra* is a vector space $\mathfrak{g}$, wiht a bilinear map

$$[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$$

called the *Lie bracket*, with

- anticommutativity $[x, y] = [-y, x]$,
- the Jacobi idenity
$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$$

Theorem 2.5. Let $G \subseteq \mathrm{GL}(n, F)$ be a matrix Lie group (where $F = \mathbb{R}$ or $\mathbb{C}$), such that log defines a chart near $I \in G$. Let

$$\mathfrak{g} = \mathrm{T}_I G$$

is identified with a (real) subspace of $\mathrm{Mat}_n(F)$. Then $\mathfrak{g}$ is a Lie algebra, with

$$[x, y] = xy - yx$$

We call $\mathfrak{g}$ the *Lie algebra of* G. We may also write

$$\mathfrak{g} = \mathrm{Lie}(G)$$

Proof. Clearly $\mathfrak{g}$ is a vector space, and $[\cdot, \cdot]$ is anticommutative and satisfies the Jacobi identity. Therefore, all we need to show is that if $x, y \in \mathfrak{g}$, then so is $[x, y]$.

For $B_1, B_2 \in \mathfrak{g}$, consider

$$A(t) = \exp(B_1 t) \exp(B_2 t) \exp(-B_1 t) \exp(-B_2 t)$$

This defines a smooth path in G, with $A(0) = I$. Recall

$$\exp(Bt) = I + Bt + \frac{1}{2} B^2 t^2 + o(t^2)$$

as $t \to 0$. We obtain that

$$A(t) = I + [B_1, B_2]t^2 + o(t^2)$$

Letting $B(t) = \log(A(t))$, for small t, we have that $B(t) = [B_1, B_2]t^2 + o(t^2)$. But we also have that

$$\exp(A(t)) = B(t)$$

Moreover, by definition, $B(t) \in \mathfrak{g}$ for t small, as $B(t)$ is in the image of the logarithm chart. So

$$\frac{B(t)}{t^2} \in \mathfrak{g}$$

as $\mathfrak{g}$ is a vector space. So

$$\lim_{t\to 0} \frac{B(t)}{t^2} = [B_1, B_2] \in \mathfrak{g}$$

since $\mathfrak{g} \leq \mathrm{Mat}_n(F)$ is closed. □

If we take $G = O(n)$, then we have that

$$\mathfrak{g} = \mathfrak{o}(n) = \left\{A \in \mathrm{Mat}_n(\mathbb{R}) \mid A^{\mathsf{T}} + A = 0\right\}$$

is the space of skew-symmetric matrices.

Definition 2.6

Let M be a manifold. Then the *tangent bundle of* M is

$$\mathrm{T}M = \bigsqcup_{p\in M} \mathrm{T}_p M$$

Theorem 2.7. $\mathrm{T}M$ has a smooth structure, making it into a manifold of dimension $2\dim(M)$.

Proof. The topology on $\mathrm{T}M$ will be induced from the smooth structure. Let (φ, U) be a chart on M. Say the local coordinates corresponding to φ are $x_1, \ldots, x_n$. So we can write $a \in \mathrm{T}_p M$ as

$$\sum a_i \frac{\partial}{\partial x_i}$$

Define

$$U_T = \bigsqcup_{p\in U} \mathrm{T}_p M$$

$$\varphi_T(p, a) = (\varphi(p), (a_i))$$

It is clear that the sets of the form U_T cover, so all we need to check is that the change of coordinate maps are smooth. Let (ψ, V) be another chart on M, with local coordinates y_i. Then

$$\psi \circ \varphi^{-1}(x, a) = (y, b)$$

and it is easy to see that $y = \psi \circ \varphi^{-1}(x)$, and

$$b_i = \sum_j \frac{\partial y_i}{\partial x_j} a_j$$

Hausdorff and second countable follow from the fact that M and $\mathbb{R}^n$ are. □

Lecture 5

We will use the notation $\pi : \mathrm{T}M \to M$ for the canonical projection map

$$\pi(p, a) = p$$

Proposition 2.8. The map $\pi : \mathrm{T}M \to M$ is smooth.

Proof. In local coordinates given by φ and φ_T, we have

$$\pi(x, a) = x$$

□

Remark 2.9. In general, the tangent bundle TM need not be diffeomorphic to $M \times \mathbb{R}^n$.

Definition 2.10 (smooth vector field)

A *smooth vector field* X on a manifold M is a smooth map $X : M \to TM$, such that $\pi(X(p)) = p$. That is, $X(p) \in T_pM$.

Example 2.11

$X = 0$ is a vector field. Similarly, X supported on a coordinate neighbourhood is a well defined vector field.

X is C^∞ if and only if for each coordinate neighbourhood U, with local coordinates x_i, we have

$$X = \sum_i a_i(p) \frac{\partial}{\partial x_i}$$

Smoothness of X becomes the requirement that each $a_i : U \to \mathbb{R}$ is smooth.

Theorem 2.12. Suppose M^n is a manifold, and $X^{(1)}, \dots, X^{(n)}$ are smooth vector fields on M, such that for all $p \in M$,

$$X^{(1)}(p), \dots, X^{(n)}(p)$$

is a basis for T_pM. Then TM is isomorphic to the product $M \times \mathbb{R}^d$. That is, there exists $\Phi : TM \to M \times \mathbb{R}^n$ is a diffeomorphism, such that

$$\Phi(p, m) = (p, \phi_p(m))$$

and for each fixed p, $\phi_p : T_pM \to \mathbb{R}^n$ is a linear isomorphism.

Manifolds which satisfies the requirements of the theorem are called *parallelisable*.

Proof. For $(p, a) \in TM$, we can write it as

$$a = \sum a_i X^{(i)}(p)$$

for some unique $a_1, \dots, a_n \in \mathbb{R}$. Using this, we can define a map

$$\Phi(p, a) = (p, (a_i)) \in M \times \mathbb{R}^n$$

It is easy to see that this is a bijection. Moreover, ϕ_p defines a linear isomorphism. Therefore, all we need to check is the smoothness. Let $\psi : U \to \mathbb{R}^n$ be a chart on M, and $\psi_T : U_T \to \mathbb{R}^n \times \mathbb{R}^n$ be the corresponding chart on TM. In this case,

$$(\psi \times \mathrm{id}) \circ \Phi \circ \psi_T^{-1}(x, b) = (x, a)$$

such that

$$a = \sum_i a_i X^{(i)}(p) = \sum_j b_j \left(\frac{\partial}{\partial x_j} \right)_p$$

We can write $X^{(i)}$ in the $\frac{\partial}{\partial x_j}$ basis, to get

$$X^{(i)} = \sum_j X_j^{(i)} \frac{\partial}{\partial x_j}$$

Thus,

$$b_j = \sum_i a_i X_j^{(i)}(x)$$

Since the $X_j^{(i)}$ are smooth, we are done. Moreover, the inverse transformation is also smooth, and thus Φ defines a local diffeomorphism, which must then be a diffeomorphism. □

Remark 2.13.

- The converse of the theorem is also true, and it is easy, since all we need to choose is (the image of) $(x, e_1), \ldots, (x, e_n)$.
- The condition that M is parallelisable is a restriction. We know that all Lie groups are parallelisable, but general manifolds are not. For example, there are no non-vanishing vector fields on S^2. More generally, S^n is parallelisable if and only if $n = 1, 3$ or 7.

Definition 2.14 (differential)

Let $F : M \to N$ be a smooth map, Define the *differential of F at $p \in M$* as

$$\begin{aligned} \mathrm{d}F_p : \mathrm{T}_p M &\to \mathrm{T}_{F(p)} N \\ \left.\frac{\partial}{\partial x_i}\right|_p &\mapsto \sum_j \frac{\partial y_i}{\partial x_j}(x(p)) \left.\frac{\partial}{\partial y_j}\right|_{F(p)} \end{aligned}$$

where x_i are local coordinates around p, given by the chart φ, and y_j are local coordinates aroung $F(p)$, given by ψ. In local coordinates, $y = y(x) = \psi \circ F \circ \psi^{-1}(x)$.

Moreover, it is easy to see this is independent of choice of local coordinate, which follows from the chain rule in multivariave calculus.

Remark 2.15. For smooth maps $F : M \to N$, $G : N \to P$,

$$\mathrm{d}(G \circ F) = \mathrm{d}G \circ \mathrm{d}F$$

Now suppose $F : M \to N$ is a diffeomorphism, X is a smooth vector field on M. Then the *pushforward of X by F* is

$$((\mathrm{d}F)X)_{F(p)} = (\mathrm{d}F)_p X(p)$$

which defines a vector field on N. We want F^{-1} to be smooth, since otherwise we don't know whether the image will be smooth.

Every vector field X on M defines a linear map

$$X : \mathbb{C}^\infty(M) \to \mathbb{C}^\infty(M)$$

where if

$$X = \sum_i X_i \frac{\partial}{\partial x_i}$$

we define locally

$$Xh = \sum_i X_i \frac{\partial h}{\partial x_i}$$

again by the chain rule, this is independent of choice of coordinates.

Let $f \in C^\infty(N)$, $F : M \to N$ be a diffeomorphism. Then $f \circ F \in C^\infty(M)$. In coordinates x_i on M, y_j on N given by F,

$$\frac{\partial}{\partial x_i}(f \circ F) = \sum_j \frac{\partial f}{\partial y_j} \frac{\partial y_j}{\partial x_i}$$

Therefore,

$$X(f \circ F) = \sum_{i,j} X_i \frac{\partial f}{\partial y_j} \frac{\partial y_j}{\partial x_i} = (((\mathrm{d}F)X)\, f) \circ F$$

That is, the diagram

$$\begin{array}{ccc} C^\infty(N) & \xrightarrow{\cdot \circ F} & C^\infty(M) \\ {\scriptstyle (\mathrm{d}F)X}\downarrow & & \downarrow{\scriptstyle X} \\ C^\infty(N) & \xrightarrow{\cdot \circ F} & C^\infty(M) \end{array}$$

commutes.

Let X, Y be vector fields on M. X and Y can be consider as a first order linear differential operator. Then the composition XY is not, but the *Lie bracket*

$$[X, Y] = XY - YX$$

is a well defined first order linear differential operator. That is, it defines an $\mathbb{R}$-linear map $C^\infty(M) \to C^\infty(M)$, satisfying

$$[X, Y](fg) = f \cdot [X, Y]g + g \cdot [X, Y]f$$

One way to see this is by the symmetry of mixed partial derivatives. Thus, the space of all vector fields on M forms a Lie algebra. Note that it is infinite dimensional, and so it can't come from a Lie group.

Lecture 6

Suppose in local coordinates $X = \sum_i X_i \frac{\partial}{\partial x_i}$ and $Y = \sum_i Y_i \frac{\partial}{\partial x_i}$, then

$$[X, Y] = \sum_{i,j} \left(X_i \frac{\partial Y_j}{\partial x_i} - Y_i \frac{\partial X_j}{\partial x_i} \right) \frac{\partial}{\partial x_k}$$

Notation 2.16. We write $V(M)$ for the vector space of all C^∞ vector fields on M.

2.1 Left invariant vector fields

Let G be a Lie group, $e \in G$ is the identity element of G, and $\mathfrak{g} = \mathrm{T}_e G$ for the tangent space of G at e, i.e. its Lie algebra.

Given $g \in G$, we can define the *left translation* $L_g : G \to G$,

$$L_g(h) = gh$$

L_g is a diffeomorphism, as it is smooth with smooth inverse $L_g^{-1} = L_{g^{-1}}$. Consider $\xi \in \mathfrak{g}$, and consider

$$X_\xi(g) = (\mathrm{d}L_g)_e \xi \in \mathrm{T}_g G$$

Since L_g is a diffeomorphism, $(\mathrm{d}L_g)_e$ is a linear isomorphism $\mathfrak{g} \to \mathrm{T}_g G$, with inverse

$$(\mathrm{d}L_g)_e^{-1} = (\mathrm{d}L_{g^{-1}})_g$$

Lemma 2.17. $X_\xi(g)$ as defined above, is smooth. Thus, X_ξ is a smooth vector field.

Proof. Consider the map $L : G \times G \to G$, with $L(g, h) = gh$. Then L is smooth. Fix $g_0 \in G$, the local coordinate expression of L in a neighbourhood of (g_0, e) is

$$\widehat{L} : U_{g_0} \times U_e \to V_{g_0}$$

Explicitly, $\widehat{L} = \varphi_{g_0} \circ L \circ (\varphi_{g_0}^{-1} \times \varphi_e^{-1})$. With this,

$$\widehat{L_g} = \widehat{L}(\varphi_{g_0}(g), \cdot) : U_e \to V_{g_0}$$

where g is in a neighbourhood of g_0. With this, consider the derivative $D_2\widehat{L}$ of $\widehat{L}$ in the V_e coordinates. Since $D_2\widehat{L}$ depends smoothly on the V_{g_0} coordinates on L, X_ξ has smooth coefficients in a neighbourhood of g_0. □

Proposition 2.18. If $\xi_1, \ldots, \xi_n \in \mathfrak{g}$ are linearly independent. That for all $g \in G$,

$$X_{\xi_1}(g), \ldots, X_{\xi_n}(g)$$

are linearly independent.

Proof. Clear from definition as $(\mathrm{d}L_g)_e$ is invertible. □

Theorem 2.19. Every Lie group is parallelisable.

Proof. From the proposition and theorem 2.12. □

Also, consider $g, h \in G$, and $\xi \in \mathfrak{g}$. In this case,

$$(\mathrm{d}L_g)_h X_\xi(h) = (\mathrm{d}L_g)_h (\mathrm{d}L_h)_e(\xi) = (\mathrm{d}L_g \circ \mathrm{d}L_h)_e(\xi) = (\mathrm{d}L_{gh})_e(\xi) = X_\xi(gh)$$

That is,

$$(\mathrm{d}L_g) X_\xi = X_\xi \circ L_g$$

Definition 2.20 (left-invariant vector field)
A vector field X on a Lie group G, with

$$(\mathrm{d}L_g) X = X \circ L_g \tag{1}$$

is called a *left-invariant vector field*. We write

$$\ell(G) \subseteq V(G)$$

for the subspace of all left-invariant vector fields on G.

It is easy to see that $\ell(G)$ is finite dimensional, since any $X \in \ell(G)$ is X_ξ for $\xi = X(e) \in \mathfrak{g}$. This also shows that the map $\mathfrak{g} \to \ell(G)$ given by $\xi \mapsto X_\xi$ is a linear isomorphism.

Thus,

$$\dim(\ell(G)) = \dim(\mathfrak{g}) = \dim(G)$$

Theorem 2.21. $\ell(G)$ is a Lie subalgebra of $V(G)$. More precisely, $[X_\xi, X_\eta]$ is left invariant for any $\xi, \eta \in \mathfrak{g}$.

Proof. Recall that

$$X(f \circ F) = (((\mathrm{d}F)X)\, f) \circ F$$

We will take $F = L_g$, $f \in C^\infty(G)$, $X = [X_\xi, X_\eta]$. We will show that X satisfies eq. (1).

$$\begin{aligned}
\big((\mathrm{d}L_g)[X_\xi, X_\eta](f)\big) \circ L_g &= [X_\xi, X_\eta](f \circ L_g) \\
&= X_\xi X_\eta (f \circ L_g) - X_\eta X_\xi (f \circ L_g) \\
&= X_\xi \cdot (\mathrm{d}L_g) X_\eta(f) \circ L_g - X_\eta \cdot (\mathrm{d}L_g) X_\xi(f) \circ L_g \\
&= ((\mathrm{d}L_g)X_\xi)((\mathrm{d}L_g)X_\eta)(f) \circ L_g - ((\mathrm{d}L_g)X_\eta)((\mathrm{d}L_g)X_\xi)(f) \circ L_g \\
&= ([(\mathrm{d}L_g)X_\xi, (\mathrm{d}L_g)X_\eta]f) \circ L_g \\
&= [X_\xi \circ L_g, X_\eta \circ L_g](f) \circ L_g \\
&= ([X_\xi, X_\eta] \circ L_g)(f) \circ L_g
\end{aligned}$$

But f, g are arbitrary, and as such, we have

$$(\mathrm{d}L_g)[X_\xi, X_\eta] = [X_\xi, X_\eta] \circ L_g$$

which means that $[X_\xi, X_\eta] \in \ell(G)$. □

With this, for all $\xi, \eta \in \mathfrak{g}$, there exists $\zeta \in \mathfrak{g}$ such that

$$[X_\xi, X_\eta] = X_\eta$$

This defines a Lie algebra structure on $\mathfrak{g}$.

Theorem 2.22. Let G be a matrix Lie group and log defines charts on G. Then the map

$$\mathfrak{g} \ni \xi \mapsto X_\xi \in \ell(G)$$

is an isomorphism of Lie algebras, where on $\mathfrak{g}$ we use the matrix commutator.

We will prove this later.

3 Submanifolds

Let M be a manifold, $N \subseteq M$ and N is a manifold. Consider the inclusion map $\iota : N \hookrightarrow M$.

Definition 3.1 (embedded submanifold)
Suppose

(i) ι is smooth,

(ii) $(\mathrm{d}\iota)_p : \mathrm{T}_p N \to \mathrm{T}_p M$ is injective for all p,

(iii) ι is a homeomorphism onto its image.

Then we say that N is an *embedded submanifold* of M.

Remark 3.2. For (iii), $D \subseteq N$ is open if and only if there exists $U \subseteq M$ open, with $D = D \cap N$.

Lecture 7

The condition (iii) is required, since we don't want an example which looks like a figure-8, or other 'weird' topological spaces, such as





Remark 3.3. If ι only satisfies (i) and (ii), we call N an *immersed submanifold*.

Remark 3.4. A slight generalisation of this is that: if N, M are manifolds, a map $\psi : N \to M$ is called an *embedding*, denoted $\psi : N \hookrightarrow M$, if ψ is injective, and satisfies conditions (i), (ii) and (iii). That is, $\psi(N) \subseteq M$ is an immersed submanifold.

If $\mathrm{d}\Psi$ is injective for all p, then we call ψ an *immersion*.

Convention: Submanifold will mean embedded submanifold unless otherwise stated.

Example 3.5 (parametrised urves and surfaces in $\mathbb{R}^3$)
In this case, condition (ii) simply means what we called a 'regular' parametrisation.

Example 3.6 (irrational twist flow)

Define the map

$$\begin{aligned}\mathbb{R} &\to T^2\\ t &\mapsto (e^{it}, e^{i\alpha t})\end{aligned}$$

In this case, we see that the map is injective and an immersion, but not an embedding as the image is dense in T^2.

Is a submanifold the same as $f^{-1}(0)$ for some smooth map $f : \mathbb{R}^n \to \mathbb{R}^m$, or more generally, $f^{-1}(p)$ for some $f : M \to N$.

In general, no. Since f is smooth, it is continuous, and so $f^{-1}(p)$ is a closed set. On the other hand,

Proposition 3.7. For every closed subset $E \subseteq \mathbb{R}^2$, we can find a smooth map $f : \mathbb{R}^2 \to \mathbb{R}$, such that $f^{-1}(0) = E$.

Proof. Omitted, optional exercise. □

Definition 3.8 (regular value)

Let $f : M \to Y$ be a smooth map between manifolds. Then $q \in Y$ is a *regular value* for f if for every $p \in f^{-1}(q)$, $(\mathrm{d}f)_p : \mathrm{T}_pM \to \mathrm{T}_qY$ is surjective.

Theorem 3.9 (preimage theorem)**.** Suppose $f : M \to Y$ is a smooth map, $y \in Y$ is a regular value, with $f^{-1}(y)$ nonempty. Then $N = f^{-1}(y)$ is an embedded submanifold of M, with

$$\dim(N) = \dim(M) - \dim(Y)$$

Proof. Omitted. □

A fact from differential topology: Suppose M is a manifold, $N \subseteq M$ a subspace (with the subspace topology). Then there is at most one smooth structure on N making it into an embedded submanifold. Therefore, it makes sense for us to say N is or is not a submanifold.

The converse of the preimage theorem is only true locally.

Proposition 3.10. Let $N \hookrightarrow M$ be an embedded submanifold, $p \in N$. Then there exists an open neighbourhood $p \in U \subseteq M$, and $f : U \to \mathbb{R}^d$, $d = \dim(M) - \dim(N)$, with $0 \in \mathbb{R}^d$ is a regular value of f, and $f^{-1}(0) = U \cap f^{-1}(0)$.

Proof. Let $\varphi : U_0 \to \mathbb{R}^n$ be a chart on M near p, with $\varphi(p) = 0$, with local coordinates $x_1, \dots, x_n$. Let $\psi : V_0 \to \mathbb{R}^\ell$ be a chart on N, $\psi(p) = 0$ and with local coordinates $u_1, \dots, u_\ell$.

We may assume without loss of generality that $V_0 = U_0 \cap N$. In this case, $\iota : N \hookrightarrow M$ is expressed locally as

$$x_i = x_i(u_1, \dots, u_\ell)$$

Thus, the rank of the Jacobian matrix

$$\left(\frac{\partial x_i}{\partial u_j}\right)$$

is ℓ, by assumption. Without loss of generality,

$$\det\left(\frac{\partial x_i}{\partial u_j}\right)_{i,j=1}^{\ell} \neq 0$$

From the inverse function theorem from multivariate calculus, we have a local inverse

$$u_i = u_i(x_1, \dots, x_\ell)$$

near 0. Moreover, for $i > \ell$, then $x_i = x_i(u) = x_i(u(x_1, \dots, x_\ell)) = h_i(x_1, \dots, x_\ell)$.

Define $f_i(x) = x_i - h_i(x_1, \dots, x_\ell)$ for $\ell < i < n$. This gives the required map, with Jacobian at 0 given in block form as $\begin{pmatrix} * & I_d \end{pmatrix}$ Thus, 0 is a regular value. □

Example 3.11

Consider

$$N = \{(x_0 : x_1 : x_2) \in \mathbb{RP}^2 \mid x_2 = 0\}$$

We can see that $N \hookrightarrow \mathbb{RP}^2$, and $N \cong S^1$. However, it is impossible to write

$$N = f^{-1}(q)$$

for all $f : \mathbb{RP}^2 \to M$ where M is a 1-manifold, $q \in M$ a regular value.

To see this, suppose we found such an f. Then consider a chart Ψ around q on M, $\psi(q) = 0$. Then we have that

$$\psi \circ f : \mathbb{RP}^2 \supseteq U \to (-1, 1)$$

with $N \subseteq U \subseteq \mathbb{RP}^2$, U is open and connected. In this case, we have

$$N = \{p \mid (\psi \circ f)(p) = 0\}$$

As q is a regular value, there exists $p_+, p_- \in U$ such that

$$\psi(f(p_+)) > 0 \quad \text{and} \quad \psi(f(p_-)) < 0$$

But $U \setminus N \subseteq \mathbb{RP}^2 \setminus N$ is connected, since it is either an annulus or a disc.

Theorem 3.12 (Whitney embedding theorem)**.** For any manifold M^n, there exists an embedding $\psi : M \hookrightarrow \mathbb{R}^{2n}$.

On the examples sheet, we will show the result for $\mathbb{R}^N$, and with the additional assumption that M is compact. It's not too difficult to show that an embedding into $\mathbb{R}^{2n+1}$ exists, Reducing to $2n$ is the main difficulty in this theorem. Moreover, this result is optimal, for example by considering

$$\mathbb{RP}^2$$

which embeds into $\mathbb{R}^4$ but not $\mathbb{R}^3$.

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